United States Patent	12398570
Kind Code	B2
Date of Patent	August 26, 2025
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Floorboard with coupling structure

Abstract

A floorboard includes a board body, a first coupling structure with a tenon and a second coupling structure with a mortise. The tenon includes a first side portion and a second side portion. A second side portion includes a first connecting surface and a second connecting surface. The mortise includes a first recess and a second recess. The second recess includes a third connecting surface and a fourth connecting surface. When two floorboards are connected, the first coupling structure of one of the two floors is matched with the second coupling structure of the other of the two floors, and the first connecting surface of one of the two floorboards is matched with the third connecting surface of the other of the two floorboards, so as to lock the connection of the two floorboards.

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Appl. No.: **18/985386**

Filed: **December 18, 2024**

Prior Publication Data

Document Identifier	Publication Date
US 20250116121 A1	Apr. 10, 2025

Foreign Application Priority Data

CN	202411527671.3	Oct. 30, 2024

Related U.S. Application Data

continuation parent-doc WO PCT/CN2024/129966 20241105 PENDING child-doc US 18985386

Publication Classification

Int. Cl.: **E04F15/02** (20060101)

U.S. Cl.:

CPC **E04F15/02038** (20130101); E04F2015/02116 (20130101); E04F2201/0138 (20130101); E04F2201/0146 (20130101); E04F2201/0161 (20130101); E04F2201/023 (20130101); E04F2201/043 (20130101)

Field of Classification Search

CPC: E04F (15/02038); E04F (2201/023); E04F (2201/043); E04F (2201/0161); E04F (2201/0138); E04F (2201/0146); E04F (2015/02116); E04F (15/02005)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of International Patent Application No. PCT/CN2024/129966, filed on Nov. 5, 2024, which claims the benefit of priority from Chinese Patent Application No. 202411527671.3, filed on Oct. 30, 2024. The content of the aforementioned application, including any intervening amendments thereto, is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) This application relates to flooring, and more particularly to a floorboard with a coupling structure.

BACKGROUND

(2) Flooring has an important functional and aesthetic value as an indoor and outdoor covering material. The spliced flooring is formed by multiple floorboards connected by a lock catch or a mortise and tenon joint structure. The spliced flooring is characterized by easy assembly, disassembly and replacement, and absence of glue. During the manufacture of the existing spliced floorboards, the mortise is usually processed to have a flat-straight edge, but such mortise fails to ensure a stable and firm connection. In the practical use, the connection will become loosened, and the gap will become wider and wider, thus affecting the service life of the flooring.

SUMMARY

(3) In view of the deficiencies in the prior art, this application provides a floorboard with improved connection strength and stability.

(4) Technical solutions of this application are described as follows

(5) This application provides a floorboard, comprising: a board body; a first coupling structure; and a second coupling structure; wherein the first coupling structure and the second coupling structure are provided on opposite sides of the board body, respectively; the first coupling structure comprises a first tenon; the second coupling structure is provided with a first mortise; and the first mortise is configured to accommodate at least a part of the first tenon for floorboard connection; the first tenon comprises a first side portion and a second side portion; the first side portion is configured to at least partially slope downwardly towards the second coupling structure; the second side portion is configured to protrude in a direction away from the second coupling structure; the second side portion comprises a first connecting surface and a second connecting surface; the first connecting surface is configured to at least partially slope downwardly from a bottom end of the first side portion towards the direction away from the second coupling structure; and the second connecting surface is configured to at least partially slope downwardly from a bottom end of the first connecting surface towards the second coupling structure; the first mortise comprises a first recess and a second recess; the first recess is located above the second recess; the first recess and the second recess are both recessed towards the first coupling structure; the second recess comprises a third connecting surface and a fourth connecting surface; the third connecting surface is configured to at least partially slope downwardly from a bottom end of the first recess towards the first coupling structure; the fourth connecting surface is configured to at least partially slope downwardly from a bottom end of the third connecting surface towards a direction away from the first coupling structure; and in response to a case of connecting two floorboards, the first coupling structure of one of the two floorboards is matched with the second coupling structure of the other of the two floorboards, and the first connecting surface of one of the two floorboards is matched with the third connecting surface of the other of the two floorboards, so as to lock the two floorboards in a connection state.

(6) In an embodiment, the first tenon further comprises a first bottom structure and a third side portion; the first bottom structure is configured to extend from an end of the second connecting surface away from the first connecting surface towards the second coupling structure; and the third side portion is configured to at least partially slope downwardly towards the direction away from the second coupling structure, and is connected to an end of the first bottom structure away from the second side portion.

(7) In an embodiment, the first mortise further comprises a first base surface and a second base surface; the first base surface is configured to extend from a bottom end of the fourth connecting surface towards the direction away from the first coupling structure; the second base surface is configured to at least partially slope downwardly towards the first coupling structure, and is

connected to an end of the first base surface away from the fourth connecting surface; and in response to a case that two floorboards are connected, the first base surface of one of the two floorboards abuts against the first bottom structure of the other of the two floorboards, and the second base surface of one of the two floorboards abuts against the third side portion of the other of the two floorboards.

(8) In an embodiment, the first bottom structure comprises a first bottom portion and a second bottom portion; an end of the first bottom portion is connected to the end of the second connecting surface away from the first connecting surface; and the first bottom portion is provided horizontally; and the second bottom portion is convex downwardly relative to the first bottom portion; and the first base surface is matched with the first bottom structure; and an area of the first base surface corresponding to the second bottom portion is downwardly recessed to fit the second bottom portion; or the second bottom portion is upwardly recessed relative to the first bottom portion and the area of the first base surface corresponding to the second bottom portion is convex upwardly to fit the second bottom portion.

(9) In an embodiment, the first coupling structure further comprises a second tenon; the second coupling structure is further provided with a second mortise; the second tenon is provided on an inner side of the first tenon; the second mortise is provided on an outer side of the first mortise; and the second mortise is configured to accommodate at least a part of the second tenon.

(10) In an embodiment, the second tenon comprises a fourth side portion, a second bottom structure and a fifth side portion; the fourth side portion is configured to at least partially extend downwardly, and is connected with a first end of the second bottom structure; the second bottom structure is configured to extend from a bottom end of the fourth side portion towards a direction away from the first tenon; and the fifth side portion is configured to at least partially extend downwardly, and is connected with a second end of the second bottom structure.

(11) In an embodiment, the fourth side portion comprises a first straight section, a first inclined section, a second straight section and a second inclined section; the first straight section is configured to extend downwardly; the first inclined section is configured to at least partially slope downwardly from an end of the first straight section towards the first tenon; the second straight section is configured to extend downwardly from an end of the first inclined section away from the first straight section; and the second inclined section is configured to at least partially slope downwardly from an end of the second straight section away from the first inclined section towards a direction away from the first tenon, and is connected to the first end of the second bottom structure.

(12) In an embodiment, the fifth side portion comprises a third inclined section, a fourth inclined section and a fifth inclined section connected in sequence; the third inclined section is configured to at least partially slope downwardly towards the first tenon, and is connected to the second end of the second bottom structure; the fourth inclined section is configured to at least partially slope downwardly towards the direction away from the first tenon, and is connected to an end of the third inclined section away from the second bottom structure; and the fifth inclined section is configured to at least partially slope downwardly towards the first tenon, and is connected to an end of the fourth inclined section away from the third inclined section.

(13) In an embodiment, the second mortise comprises a first plane, a second plane, a third plane, a fourth plane, a fifth plane, a sixth plane, a seventh plane and an eighth plane connected in sequence; the first plane is configured to extend downwardly; the second plane is configured to at least partially slope downwardly from an end of the first plane towards the first mortise; the third plane is configured to extend downwardly from an end of the second plane away from the first plane; the fourth plane is configured to at least partially slope downwardly from an end of the third plane away from the second plane towards a direction away from the first mortise; the fifth plane is configured to extend from an end of the fourth plane away from the third plane towards the direction away from the first mortise; the sixth plane is configured to slope downwardly towards

the first mortise, and is connected to an end of the fifth plane away from the fourth plane; the seventh plane is configured to at least partially slope downwardly towards the direction away from the first mortise, and is connected to an end of the sixth plane away from the fifth plane; and the eighth plane is configured to at least partially slope downwardly towards the first mortise, and is connected to an end of the seventh plane away from the sixth plane; and in response to a case that two floorboards are connected, the first plane of one of the two floorboards abuts against the first straight section of the other of the two floorboards, the second plane of one of the two floorboards abuts against the first inclined section of the other of the two floorboards, the third plane of one of the two floorboards abuts against the second straight section of the other of the two floorboards, the fourth plane of one of the two floorboards abuts against the second inclined section of the other of the two floorboards, the fifth plane of one of the two floorboards abuts against the second bottom structure of the other of the two floorboards, the sixth plane of one of the two floorboards abuts against the third inclined section of the other of the two floorboards, the seventh plane of one of the two floorboards abuts against the fourth inclined section of the other of the two floorboards, and the eighth plane of one of the two floorboards abuts against the fifth inclined section of the other of the two floorboards.

(14) In an embodiment, the second bottom structure comprises a first bottom portion and a second bottom portion; a first end of the first bottom portion is connected to an end of the second inclined section away from the second straight section, and a second end of the first bottom portion is connected to a first end of the second bottom portion; a second end of the second bottom portion is connected to an end of the third inclined section away from the fourth inclined section; and the first bottom portion is provided horizontally; and the second bottom portion is recessed upwardly relative to the first bottom portion; the fifth plane is matched with the second bottom structure; and an area of the fifth plane corresponding to the second bottom portion is convex upwardly to fit the second bottom portion; or the second bottom portion is convex downwardly relative to the first bottom portion; and the area of the fifth plane corresponding to the second bottom portion is recessed downwardly to fit the second bottom portion.

(15) In an embodiment, a third mortise is provided between the first tenon and the second tenon; a third tenon is provided between the first mortise and the second mortise; and the third tenon is configured to fit the third mortise; and a sixth inclined section is provided between the third side portion and the fourth side portion; the sixth inclined section is configured to at least partially slope downwardly from an end of the third side portion away from the first bottom structure towards the direction away from the first tenon, and is connected to an end of the fourth side portion away from the second bottom structure; and the third side portion, the sixth inclined section and the fourth side portion are configured to together enclose the third mortise.

(16) In an embodiment, the second coupling structure further comprises a ninth plane; the ninth plane is configured to slope downwardly from an end of the second base surface away from the first base surface towards the direction away from the first mortise, and is connected to an end of the first plane away from the second plane; the second base surface, the ninth plane, the first plane, the second plane, the third plane and the fourth plane are configured to together enclose the third tenon; and in response to a case that two floorboards are connected, the second base surface of one of the two floorboards abuts against the third side portion of the other of the two floorboards, the ninth plane of one of the two floorboards abuts against the sixth inclined section of the other of the two floorboards, the first plane of one of the two floorboards abuts against the first straight section of the other of the two floorboards, the second plane of one of the two floorboards abuts against the first inclined section of the other of the two floorboards, the third plane of one of the two floorboards abuts against the second straight section of the other of the two floorboards, and the fourth plane of one of the two floorboards abuts against the second inclined section of the other of the two floorboards.

(17) Compared with the prior art, this application has the following technical effects.

(18) A floorboard provided in this application includes a board body, and a first coupling structure and a second coupling structure provided on opposite sides of the board body, respectively. A first tenon of the first coupling structure includes a first side portion and a second side portion. A first mortise of the second coupling structure includes a first recess and a second recess. When two floorboards are connected, the first side portion of one of the two floorboards is partially inserted into the first recess of the other of the two floorboards, and the second side portion of one of the two floorboards is partially inserted into the second recess of the other of the two floorboards. The first connecting surface of the second side portion and the third connecting surface of the second recess are inclined relative to the board body, so when two floorboards are connected, the first connecting surface of one of the two floorboards abuts against the third connecting surface of the other of the two floorboards to lock the floorboards in the horizontal direction and the vertical direction, thereby enhancing the connection stability and reliability and enhancing the connection strength of the floorboards.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to illustrate technical solutions in embodiments of the present disclosure or the prior art more clearly, the required drawings will be briefly described below. Obviously, presented in the drawings are merely some embodiments of the present disclosure, which are not intended to limit the disclosure. For those skilled in the art, other drawings may also be obtained according to the drawings provided herein without paying creative efforts.

(2) FIG. **1** is a schematic diagram of a floorboard according to Embodiment 1 of the present disclosure;

(3) FIG. **2** schematically shows connection of two floorboards according to Embodiment 1 of the present disclosure;

(4) FIG. **3** is a front view of the two floorboards according to Embodiment 1 of the present disclosure in a connection state;

(5) FIG. **4** is an enlarged view of portion “A” in FIG. **3**;

(6) FIG. **5** is a front view of the floorboard according to Embodiment 1 of the present disclosure;

(7) FIG. **6** is an enlarged view of portion “B” in FIG. **5**;

(8) FIG. **7** is an enlarged view of portion “D” in FIG. **6**;

(9) FIG. **8** is an enlarged view of portion “C” in FIG. **5**;

(10) FIG. **9** is an enlarged view of portion “E” in FIG. **8**;

(11) FIG. **10** schematically shows connection of two floorboards according to Embodiment 2 of the present disclosure;

(12) FIG. **11** is a front view of the two floorboards according to Embodiment 2 of the present disclosure in a connection state;

(13) FIG. **12** is an enlarged view of portion “F” in FIG. **11**;

(14) FIG. **13** schematically shows connection of two floorboards according to Embodiment 3 of the present disclosure;

(15) FIG. **14** is a front view of the two floorboards according to Embodiment 3 of the present disclosure in a connection state;

(16) FIG. **15** is an enlarged view of portion “G” in FIG. **14**;

(17) FIG. **16** schematically shows connection of two floorboards according to Embodiment 4 of the present disclosure;

(18) FIG. **17** is a front view of the two floorboards according to Embodiment 4 of the present disclosure in a connection state; and

(19) FIG. **18** is an enlarged view of portion “H” in FIG. **17**.

(20) **1**, board body; **2**, first coupling structure; **3**, second coupling structure; **4**, first tenon; **5**, first mortise; **6**, first side portion; **7**, second side portion; **8**, first bottom structure; **9**, third side portion; **10**, first connecting surface; **11**, second connecting surface; **12**, first recess; **13**, second recess; **14**, third connecting surface; **15**, fourth connecting surface; **16**, first base surface; **17**, second base surface; **18**, first cavity; **19**, second cavity; **20**, fifth connecting surface; **21**, sixth connecting surface; **22**, second tenon; **23**, second mortise; **24**, fourth side portion; **25**, second bottom structure; **26**, fifth side portion; **27**, first straight section; **28**, first inclined section; **29**, second straight section; **30**, second inclined section; **31**, third inclined section; **32**, fourth inclined section; **33**, fifth inclined section; **34**, first plane; **35**, second plane; **36**, third plane; **37**, fourth plane; **38**, fifth plane; **39**, sixth plane; **40**, seventh plane; **41**, eighth plane; **42**, third mortise; **43**, third tenon; **44**, sixth inclined section; **45**, ninth plane; **46**, fourth mortise; **47**, fourth straight section; **48**, fifth straight section; **49**, fourth tenon; **50**, tenth plane; **51**, eleventh plane; **52**, first bottom portion; **53**, second bottom portion; **54**, third bottom portion; **55**, fourth bottom portion; and **56**, top surface.
(21) The present disclosure will be further described in detail below in conjunction with the accompanying drawings and embodiments to make the objects, features and advantages of the present disclosure better understood.

DETAILED DESCRIPTION OF EMBODIMENTS

(22) The technical solutions of the disclosure will be clearly and completely described in detail below in combination with the drawings in the embodiments. Obviously, described below are merely some embodiments of the disclosure, which are not intended to limit the disclosure. For those skilled in the art, other embodiments obtained based on these embodiments without paying creative efforts should fall within the scope of the disclosure defined by the appended claims.
(23) As used herein, it should be noted that the orientation or positional relationship indicated by the terms “up”, “down”, “left”, “right”, “front”, and “back”, is merely used to describe the relative orientation or positional relationship or movement between individual components in a particular attitude (e.g., in the accompanying drawings), and when the particular attitude is changed, the orientation indication changes accordingly.
(24) In addition, the terms “first”, “second” and the like are merely descriptive, and cannot be understood as indicating or implying relative importance, or implying the number of indicated technical features. The terms “first” and “second” indicate at least one of such feature. As used herein, the term “and/or” refers to any possible combinations of one or more of the items listed, and includes those combinations. For example, A and/or B includes A, a combination of A and B, and B. Furthermore, the technical solutions of various embodiments may be combined with each other in the case of no contradiction.

Embodiment 1

(25) Referring to FIGS. **1-9**, a floorboard includes a board body **1**, a first coupling structure **2** and a second coupling structure **3**. The first coupling structure **2** and the second coupling structure **3** are disposed on opposite sides of the board body **1**, respectively. The first coupling structure **2** includes a first tenon **4**, and the second coupling structure **3** is provided with a first mortise **5**. The first mortise **5** is used to accommodate at least a part of the first tenon **4** for floorboard connection.
(26) The first tenon **4** includes a first side portion **6**, a second side portion **7**, a first bottom structure **8**, and a third side portion **9**. The first side portion **6** at least partially slopes downwardly towards a direction close to the second coupling structure **3**. The second side portion **7** protrudes in a direction away from the second coupling structure **3**. The second side portion **7** includes a first connecting surface **10** and a second connecting surface **11**. The first connecting surface **10** at least partially slopes downwardly from the bottom end of the first side portion **6** towards the direction away from the second coupling structure **3**. The second connecting surface **11** at least partially slopes downwardly from the bottom end of the first connecting surface **10** in a direction close to the second coupling structure **3**. The first bottom structure **8** is configured to extend from the end of the second connecting surface **11** away from the first connecting surface **10** towards a direction

close to the second coupling structure **3**. The third side portion **9** at least partially slopes downwardly towards the direction away from the second coupling structure **3** and is connected to an end of the first bottom structure **8** away from the second side portion **7**.

(27) The first mortise **5** includes a first recess **12** and a second recess **13**. The first recess **12** is disposed above the second recess **13**. The first recess **12** and the second recess **13** are both recessed towards the direction close to the first coupling structure **2**. The second recess **13** includes a third connecting surface **14** and a fourth connecting surface **15**, and the first mortise **5** further includes a first base surface **16** and a second base surface **17**. The third connecting surface **14** at least partially slopes downwardly from the bottom end of the first recess **12** towards the direction close to the first coupling structure **2**. The fourth connecting surface **15** at least partially slopes downwardly from the bottom end of the third connecting surface **14** towards a direction away from the first coupling structure **2**. The first base surface **16** is configured to extend from a bottom end of the fourth connecting surface **15** towards the direction away from the first coupling structure **2**. The second base surface **17** is configured to at least partially slope downwardly towards a direction close to the first coupling structure **2** and is connected to the end of the first base surface **16** away from the fourth connecting surface **15**.

(28) In this embodiment, when the floorboards are in the connection state, the first side portion **6** and the first recess **12** enclose a first cavity **18**. The first connecting surface **10** and the third connecting surface **14** abuts to each other to lock the floorboards in both the horizontal direction and the vertical direction in the connection state. The first connecting surface **10**, the second connecting surface **11**, the third connecting surface **14** and the fourth connecting surface **15** are configured to enclose a second cavity **19**. The first base surface **16** abuts against the first bottom structure **8** to lock the floorboards in the vertical direction for coupling the floorboards. The second base surface **17** abuts against the third side portion **9** to further provide force components in the horizontal and vertical directions for the floorboard connection to improve the connection stability.

(29) Since there is dimensional error when the floorboard is grooved, the first cavity **18** and the second cavity **19** can reduce the phenomenon of uneven upper parts when two floorboards are joined due to dimensional errors caused by grooving.

(30) The first bottom structure **8** is horizontally provided and is parallel to the width direction of the board body **1**.

(31) Specifically, the horizontal direction is parallel to the width direction of the board body **1**, and the vertical direction is parallel to the thickness direction of the board body **1**.

(32) Specifically, the first connecting surface **10** and the second connecting surface **11** are transitionally connected by a circular arc. The third connecting surface **14** and the fourth connecting surface **15** are transitionally connected by a circular arc. The curved surface is designed for reducing the friction between the first tenon **4** and the first mortise **5** when spliced together, so as to make the splicing smoother and avoid jamming.

(33) In this embodiment, the first coupling structure **2** further includes a fifth connecting surface **20**. The second coupling structure **3** further includes a sixth connecting surface **21**. The fifth connecting surface **20** is disposed vertically or tilted above the first side portion **6** and connected to the top surface of the floorboard. The sixth connecting surface **21** is disposed vertically or tilted above the first recess **12** and connected to the top surface of the floorboard. The fifth connecting surface **20** and the sixth connecting surface **21** are inclined at equal angles with respect to the thickness direction of the board body **1** to achieve seamless fitting of the fifth connecting surface **20** and the sixth connecting surface **21**. Further, there is a certain distance between the first cavity **18** and the top surface **56** of the floorboard. In response to a case of connecting two floorboards, the first coupling structure **2** of one of the two floorboards can be tightly affixed to the upper portion of the side surface of the second coupling structure **3** of the other of the two floorboards. The fifth connecting surface **20** of the first coupling structure **2** and the sixth connecting surface **21** of the second coupling structure **3** are tightly affixed to each other, which not only meets the demand for

aesthetics but also avoids debris from entering the first cavity **18**.
(34) Referring to FIGS. **6** and **8**, the first coupling structure **2** further includes a second tenon **22**. The second coupling structure **3** is further provided with a second mortise **23**. The second tenon **22** is provided on the inner side of the first tenon **4**. The second mortise **23** is provided on the outer side of the first mortise **5**. The second mortise **23** is used to accommodate at least a part of the second tenon **22**.
(35) Referring to FIGS. **6** and **7**, the second tenon **22** includes a fourth side portion **24**, a second bottom structure **25**, and a fifth side portion **26**. The fourth side portion **24** is configured to at least partially extend downwardly and is connected to one end of the second bottom structure **25**. The second bottom structure **25** is configured to extend from the bottom end of the fourth side portion **24** towards a direction away from the first tenon **4**. The fifth side portion **26** is configured to at least partially extend downwardly and is connected to the other end of the second bottom structure **25**.
(36) Specifically, the fourth side portion **24** includes a first straight section **27**, a first inclined section **28**, a second straight section **29** and a second inclined section **30**. The first straight section **27** is configured to extend downwardly. The first inclined section **28** is configured to at least partially slope downwardly from one end of the first straight section **27** towards the direction close to the first tenon **4**. The second straight section **29** is configured to extend downwardly from one end of the first inclined section **28** away from the first straight section **27**. The second inclined section **30** is configured to at least partially slope downwardly from the end of the second straight section **29** away from the first inclined section **28** towards a direction away from the first tenon **4**, and is connected to an end of a second bottom structure **25**. The second bottom structure **25** is configured to extend from the end of the second inclined section **30** away from the second straight section **29** towards the direction away from the first tenon **4**.
(37) The fifth side portion **26** includes a third inclined section **31**, a fourth inclined section **32** and a fifth inclined section **33** which are connected to each other. The third inclined section **31** at least partially slopes downwardly towards a direction close to the first tenon **4** and is connected to the end of the second bottom structure **25** away from the second inclined section **30**. The fourth inclined section **32** at least partially slopes downwardly towards the direction away from the first tenon **4** and is connected to the end of the third inclined section **31** away from the second bottom structure **25**. The fifth inclined section **33** at least partially slope downwardly towards a direction close to the first tenon **4** and is connected to the end of the fourth inclined section **32** away from the third inclined section **31**.
(38) Specifically, the second bottom structure **25** includes a third straight section which is substantially flat.
(39) Referring to FIGS. **8** and **9**, the second mortise **23** includes a first plane **34**, a second plane **35**, a third plane **36**, a fourth plane **37**, a fifth plane **38**, a sixth plane **39**, a seventh plane **40**, and an eighth plane **41** connected in sequence. The first plane **34** is configured to extend downwardly. The second plane **35** at least partially slopes downwardly from the end of the first plane **34** towards the direction close to the first mortise **5**. The third plane **36** is configured to extend downwardly from an end of the second plane **35** away from the first plane **34**. The fourth plane **37** is configured to at least partially slope downwardly from an end of the third plane **36** away from the second plane **35** towards the direction away from the first mortise **5**. The fifth plane **38** is configured to extend downwardly from an end of the fourth plane **37** away from the third plane **36** towards the direction away from the first mortise **5**. The sixth plane **39** slopes downwardly towards the direction close to the first mortise **5**, and is connected to the end of the fifth plane **38** away from the fourth plane **37**. The seventh plane **40** at least partially downwardly slopes towards the direction away from the first mortise **5**, and is connected to the end of the sixth plane **39** away from the fifth plane **38**. The eighth plane **41** at least partially slopes downwardly in the direction close to the first mortise **5**, and is connected to the end of the seventh plane **40** away from the sixth plane **39**.
(40) In response to a case that two floorboards are connected, the first plane **34** of one of the two

floorboards abuts against the first straight section **27** of the other of the two floorboards; the second plane **35** of one of the two floorboards abuts against the first inclined section **28** of the other of the two floorboards; the third plane **36** of one of the two floorboards abuts against the second straight section **29** of the other of the two floorboards; the fourth plane **37** of one of the two floorboards abuts against the second inclined section **30** of the other of the two floorboards; the fifth plane **38** of one of the two floorboards abuts against the second bottom structure **25** of the other of the two floorboards; the sixth plane **39** of one of the two floorboards abuts against the third inclined section **31** of the other of the two floorboards; the seventh plane **40** of one of the two floorboards abuts against the fourth inclined section **32** of the other of the two floorboards; and the eighth plane **41** of one of the two floorboards abuts against the fifth inclined section **33** of the other of the two floorboards, further improving the connection stability of the two floorboards.
(41) As shown in FIG. **5**, when the floorboards are installed, the first mortise **5** and the second mortise **23** faces upwards, and the first tenon **4** and the second tenon **22** faces downwards.
(42) In this embodiment, the vertical distance between the first bottom structure **8** and the top surface **56** of the floorboard is greater than the vertical distance between the second bottom structure **25** and the top surface **56** of the floorboard, which facilitates the insertion of the first tenon **4** into the first mortise **5** during installation.
(43) Referring to FIGS. **3** and **6-9**, a third mortise **42** is provided between the first tenon **4** and the second tenon **22**. A third tenon **43** is provided between the first mortise **5** and the second mortise **23**. The third tenon **43** is used to fit the third mortise **42**. A sixth inclined section **44** is provided between the third side portion **9** and the fourth side portion **24**. The sixth inclined section **44** at least partially slopes downwardly from an end of the third side portion **9** away from the first bottom structure **8** towards the direction away from the first tenon **4**, and is connected to an end of the fourth side portion **24** away from the second bottom structure **25**. The third side portion **9**, the sixth inclined section **44**, and the fourth side portion **24** are configured to together enclose the third mortise **42**.
(44) Specifically, the second coupling structure **3** further includes a ninth plane **45**. The ninth plane **45** slopes downwardly from the end of the second base surface **17** away from the first base surface **16** towards the direction away from the first mortise **5**, and is connected to the end of the first plane **34** away from the second plane **35**. The second base surface **17**, the ninth plane **45**, the first plane **34**, the second plane **35**, the third plane **36** and the fourth plane **37** enclose the third tenon **43**. When the two floorboards are connected, the second base surface **17** of one of the two floorboards abuts against the third side portion **9** of the other of the two floorboards; the ninth plane **45** of one of the two floorboards abuts against the sixth inclined section **44** of the other of the two floorboards; the first plane **34** of one of the two floorboards abuts against the first straight section **27** of the other of the two floorboards; the second plane **35** of one of the two floorboards abuts against the first inclined section **28** of the other of the two floorboards; the third plane **36** of one of the two floorboards abuts against the second straight section **29** of the other of the two floorboards; and the fourth plane **37** of one of the two floorboards abuts against the second inclined section **30** of the other of the two floorboards.
(45) Further, the first coupling structure **2** is further provided with a fourth mortise **46**. The first coupling structure **2** further includes a fourth straight section **47** and a fifth straight section **48**. The fourth straight section **47** is configured to extend from the end of the fifth inclined section **33** away from the fourth inclined section **32** towards the direction away from the first tenon **4**. The fifth straight section **48** is configured to extend downwardly from the end of the fifth plane **38** away from the fifth inclined section **33**. The third inclined section **31**, the fourth inclined section **32**, the fifth inclined section **33**, the fourth straight section **47**, and the fifth straight section **48** enclose the fourth mortise **46**.
(46) In this embodiment, the fourth straight section **47** is configured to extend horizontally. The fifth straight section **48** is configured to extend vertically. In other embodiments, the fifth straight

section **48** slopes with respect to the board body **1**, and the fifth straight section **48** slopes at an angle of 30-60 degrees with respect to the thickness direction of the board body **1**, and preferably at the angle of 40-50 degrees.

(47) Accordingly, the second coupling structure **3** further includes a fourth tenon **49**. The second coupling structure **3** further includes a tenth plane **50** and an eleventh plane **51**. The tenth plane **50** is configured to extend from an end of the eighth plane **41** away from the seventh plane **40** towards a direction away from the first mortise **5**. The eleventh plane **51** is configured to extend downwardly from an end of the tenth plane **50** away from the eighth plane **41**. The sixth plane **39**, the seventh plane **40**, the eighth plane **41**, the tenth plane **50** and the eleventh plane **51** are configured to enclose the fourth tenon **49**. The fourth tenon **49** of one of the two floorboards is inserted into the fourth mortise **46** of the other of the two floorboards. The sixth plane **39** of one of the two floorboards abuts against the third inclined section **31** of the other of the two floorboards. The seventh plane **40** of one of the two floorboards abuts against the fourth inclined section **32** of the other of the two floorboards. The eighth plane **41** of one of the two floorboards abuts against the fifth inclined section **33** of the other of the two floorboards. The tenth plane **50** of one of the two floorboards abuts against the fourth straight section **47** of the other of the two floorboards. The eleventh plane **51** of one of the two floorboards abuts against the fifth straight section **48** of the other of the two floorboards.

(48) In this embodiment, the tenth plane **50** is configured to extend horizontally. The eleventh plane **51** is configured to extend vertically. In other embodiments, the eleventh plane **51** slopes with respect to the board body **1**. The eleventh plane **51** and the fifth straight section **48** have the same inclined angle. The inclined angle of the eleventh plane **51** with respect to the thickness direction of the board body **1** is 30-60 degrees, preferably 40-50 degrees.

(49) In this disclosure, the first tenon **4** is embedded in the first mortise **5**, the second tenon **22** is embedded in the second mortise **23**, the third tenon **43** is embedded in the third mortise **42**, and the fourth tenon **49** is embedded in the fourth mortise **46**. The multiple tenons and mortises are combined to enhance the strength of the floor joints and improve the stability of the floor joints. The multiple tenons and mortises make the force more evenly distributed and reduce the concentration of stress in one place, reducing the risk of deformation and damage to the floor material. The tenons are inserted into the mortises with a plurality of contact surfaces against each other, thereby providing stability for the floor joints when subjected to horizontal or vertical forces. Furthermore, a plurality of contact surfaces share the forces simultaneously, improving the service life and reliability of the first coupling structure **22** and the second coupling structure **33**.

(50) The installation process of two floorboards are as follows. The first tenon **4** of one of the two floorboards is first inserted into the first mortise **5** of the other of the two floorboards during installation. A force is applied to the top of the second tenon **22** to press the floor down, the second tenon **22** of one of the two floorboards is inserted into the second mortise **23** of the other of the two floorboards, at the same time, the third tenon **43** of one of the two floorboards is inserted into the third mortise **42** of the other of the two floorboards, and the fourth tenon **49** of one of the two floorboards is inserted into the fourth mortise **46** of the other of the two floorboards. The fifth connecting surface **20** of the first coupling structure **2** of one of the two floorboards is tightly affixed to the sixth connecting surface **21** of the second coupling structure **3** of the other of the two floorboards. The eleventh plane **51** of one of the two floorboards is next to the fifth straight section **48** of the other of the two floorboards.

Embodiment 2

(51) Referring to FIGS. **10-12**, differences from embodiment 1 are as follows. The first bottom structure **8** includes a first bottom portion **52** and a second bottom portion **53**. One end of the first bottom portion **52** is connected to an end of the second connecting surface **11** away from the first connecting surface **10**, and the other end of the first bottom portion **52** is connected to an end of the second bottom portion **53**. The other end of the second bottom portion **53** is connected to an end of

the third side portion **9**. The first bottom portion **52** is disposed horizontally. The second bottom portion **53** is convex downwardly with respect to the first bottom portion **52**. Correspondingly, the first base surface **16** is provided to match the first bottom structure **8**. The area of the first base surface **16** corresponding to the second bottom portion **53** is downwardly recessed to fit the second bottom portion **53**. The second bottom structure **25** includes a third bottom portion **54** and a fourth bottom portion **55**. One end of the third bottom portion **54** is connected to an end of the second inclined section **30** away from the second straight section **29**, and the other end of the third bottom portion **54** is connected to an end of the fourth bottom portion **55**. The other end of the fourth bottom portion **55** is connected to the end of the third inclined section **31** away from the fourth inclined section **32**. The third bottom portion **54** is provided horizontally. The fourth bottom portion **55** is recessed upwardly with respect to the third bottom portion **54**. Correspondingly, the fifth plane **38** of the second mortise **23** is matched with the second bottom structure **25**. The area of the fifth plane **38** corresponding to the fourth bottom portion **55** is convex upwardly to fit the fourth bottom portion **55**.
(52) In other embodiments, the second bottom portion **53** is convex downwardly with respect to the first bottom portion **52**. The fourth bottom portion **55** is convex downwardly with respect to the third bottom portion **54**. The position of the fifth plane **38** corresponding to the fourth bottom portion **55** is recessed downwardly.
Embodiment 3
(53) Referring to FIGS. **13-15**, differences from embodiment 2 are as follows. The second bottom portion **53** of the first bottom structure **8** is upwardly recessed. The area of the first base surface **16** corresponding to the second bottom portion **53** is convex upwardly. The fourth bottom portion **55** of the second bottom structure **25** is convex downwardly relative to the third bottom portion **54**. The area of the fifth plane **38** of the second mortise **23** corresponding to the fourth bottom portion **55** is downwardly recessed.
(54) In other embodiments, the second bottom portion **53** of the first bottom structure **8** is recessed upwardly. The area of the first base surface **16** corresponding to the second bottom portion **53** is convex upwardly. The fourth bottom portion **55** of the second bottom structure **25** is recessed upwardly with respect to the third bottom portion **54**. The area of the fifth plane **38** of the second mortise **23** corresponding to the fourth bottom portion **55** is convex upwardly.
Embodiment 4
(55) Referring to FIGS. **16-18**, differences from embodiment 1 are as follows. The first side portion **6** is completely embedded with the first recess **12**. The second side portion **7** is completely embedded with the second recess **13**. The first cavity **18** and the second cavity **19** are omitted. The second tenon **22** is embedded with the second mortise **23**. The fourth side portion **24** and the fifth side portion **26** are curved. The fourth side portion **24** is recessed in a direction close to the first tenon **4**, and the fifth side portion **26** is recessed in a direction away from the first tenon **4**.
(56) Described above are merely preferred embodiments of the disclosure, which are not intended to limit the disclosure. It should be understood that any modifications and replacements made by those skilled in the art without departing from the spirit of the disclosure should fall within the scope of the disclosure defined by the appended claims.

Claims

1. A floorboard, comprising: a board body; a first coupling structure; and a second coupling structure; wherein the first coupling structure and the second coupling structure are provided on opposite sides of the board body, respectively; the first coupling structure comprises a first tenon; the second coupling structure is provided with a first mortise; and the first mortise is configured to accommodate at least a part of the first tenon for floorboard connection; the first tenon comprises a first side portion and a second side portion; the first side portion is configured to at least partially

slope downwardly towards the second coupling structure; the second side portion is configured to protrude in a direction away from the second coupling structure; the second side portion comprises a first connecting surface and a second connecting surface; the first connecting surface is configured to at least partially slope downwardly from a bottom end of the first side portion towards the direction away from the second coupling structure to form a cambered surface; the second connecting surface is configured to at least partially slope downwardly from a bottom end of the first connecting surface towards the second coupling structure to form a cambered surface; and the first connecting surface is connected to the second connecting surface; the first mortise comprises a first recess and a second recess; the first recess is located above the second recess; the first recess and the second recess each extend in a direction towards the first coupling structure; the second recess comprises a third connecting surface and a fourth connecting surface; the third connecting surface is configured to at least partially slope downwardly from a bottom end of the first recess towards the first coupling structure; the fourth connecting surface is configured to at least partially slope downwardly from a bottom end of the third connecting surface towards a direction away from the first coupling structure; the first mortise further comprises a first base surface; and the first base surface is configured to extend from a bottom end of the fourth connecting surface; and in response to a case of connecting two floorboards, the first coupling structure of one of the two floorboards is matched with the second coupling structure of the other of the two floorboards, and the first connecting surface of one of the two floorboards is matched with the third connecting surface of the other of the two floorboards, so as to lock the two floorboards in a connection state; the first tenon further comprises a first bottom structure and a third side portion; the first bottom structure is configured to extend from an end of the second connecting surface away from the first connecting surface towards the second coupling structure; and the third side portion is configured to at least partially slope downwardly towards the direction away from the second coupling structure, and is connected to an end of the first bottom structure away from the second side portion; the first mortise further comprises a second base surface; the second base surface is configured to at least partially slope downwardly towards the first coupling structure, and is connected to an end of the first base surface away from the fourth connecting surface; and in response to a case that two floorboards are connected, the first base surface of one of the two floorboards abuts against the first bottom structure of the other of the two floorboards, and the second base surface of one of the two floorboards abuts against the third side portion of the other of the two floorboards; and the first bottom structure comprises a first bottom portion and a second bottom portion; an end of the first bottom portion is connected to the end of the second connecting surface away from the first connecting surface; and the first bottom portion is provided horizontally; and the second bottom portion is convex downwardly relative to the first bottom portion; the first base surface is matched with the first bottom structure; and an area of the first base surface corresponding to the second bottom portion is downwardly recessed to fit the second bottom portion; or the second bottom portion is upwardly recessed relative to the first bottom portion and the area of the first base surface corresponding to the second bottom portion is convex upwardly to fit the second bottom portion.

2. The floorboard of claim 1, wherein the first coupling structure further comprises a second tenon; the second coupling structure is further provided with a second mortise; the second tenon is provided on an inner side of the first tenon; the second mortise is provided on an outer side of the first mortise; and the second mortise is configured to accommodate at least a part of the second tenon.

3. The floorboard of claim 2, wherein the second tenon comprises a fourth side portion, a second bottom structure and a fifth side portion; the fourth side portion is configured to at least partially extend downwardly, and is connected with a first end of the second bottom structure; the second bottom structure is configured to extend from a bottom end of the fourth side portion towards a direction away from the first tenon; and the fifth side portion is configured to at least partially

extend downwardly, and is connected with a second end of the second bottom structure.
4. The floorboard of claim 3, wherein the fourth side portion comprises a first straight section, a first inclined section, a second straight section and a second inclined section; the first straight section is configured to extend downwardly; the first inclined section is configured to at least partially slope downwardly from an end of the first straight section towards the first tenon; the second straight section is configured to extend downwardly from an end of the first inclined section away from the first straight section; and the second inclined section is configured to at least partially slope downwardly from an end of the second straight section away from the first inclined section towards a direction away from the first tenon, and is connected to the first end of the second bottom structure.
5. The floorboard of claim 4, wherein the fifth side portion comprises a third inclined section, a fourth inclined section and a fifth inclined section connected in sequence; the third inclined section is configured to at least partially slope downwardly towards the first tenon, and is connected to the second end of the second bottom structure; the fourth inclined section is configured to at least partially slope downwardly towards the direction away from the first tenon, and is connected to an end of the third inclined section away from the second bottom structure; and the fifth inclined section is configured to at least partially slope downwardly towards the first tenon, and is connected to an end of the fourth inclined section away from the third inclined section.
6. The floorboard of claim 5, wherein the second mortise comprises a first plane, a second plane, a third plane, a fourth plane, a fifth plane, a sixth plane, a seventh plane and an eighth plane connected in sequence; the first plane is configured to extend downwardly; the second plane is configured to at least partially slope downwardly from an end of the first plane towards the first mortise; the third plane is configured to extend downwardly from an end of the second plane away from the first plane; the fourth plane is configured to at least partially slope downwardly from an end of the third plane away from the second plane towards a direction away from the first mortise; the fifth plane is configured to extend from an end of the fourth plane away from the third plane towards the direction away from the first mortise; the sixth plane is configured to slope downwardly towards the first mortise, and is connected to an end of the fifth plane away from the fourth plane; the seventh plane is configured to at least partially slope downwardly towards the direction away from the first mortise, and is connected to an end of the sixth plane away from the fifth plane; and the eighth plane is configured to at least partially slope downwardly towards the first mortise, and is connected to an end of the seventh plane away from the sixth plane; and in response to a case that two floorboards are connected, the first plane of one of the two floorboards abuts against the first straight section of the other of the two floorboards, the second plane of one of the two floorboards abuts against the first inclined section of the other of the two floorboards, the third plane of one of the two floorboards abuts against the second straight section of the other of the two floorboards, the fourth plane of one of the two floorboards abuts against the second inclined section of the other of the two floorboards, the fifth plane of one of the two floorboards abuts against the second bottom structure of the other of the two floorboards, the sixth plane of one of the two floorboards abuts against the third inclined section of the other of the two floorboards, the seventh plane of one of the two floorboards abuts against the fourth inclined section of the other of the two floorboards, and the eighth plane of one of the two floorboards abuts against the fifth inclined section of the other of the two floorboards.
7. The floorboard of claim 6, wherein the second bottom structure comprises a first bottom portion and a second bottom portion; a first end of the first bottom portion is connected to an end of the second inclined section away from the second straight section, and a second end of the first bottom portion is connected to a first end of the second bottom portion; a second end of the second bottom portion is connected to an end of the third inclined section away from the fourth inclined section; and the first bottom portion is provided horizontally; and the second bottom portion is recessed upwardly relative to the first bottom portion; the fifth plane is matched with the second bottom

structure; and an area of the fifth plane corresponding to the second bottom portion is convex upwardly to fit the second bottom portion; or the second bottom portion is convex downwardly relative to the first bottom portion; and the area of the fifth plane corresponding to the second bottom portion is recessed downwardly to fit the second bottom portion.

8. The floorboard of claim 6, wherein a third mortise is provided between the first tenon and the second tenon; a third tenon is provided between the first mortise and the second mortise; and the third tenon is configured to fit the third mortise; and a sixth inclined section is provided between the third side portion and the fourth side portion; the sixth inclined section is configured to at least partially slope downwardly from an end of the third side portion away from the first bottom structure towards the direction away from the first tenon, and is connected to an end of the fourth side portion away from the second bottom structure; and the third side portion, the sixth inclined section and the fourth side portion are configured to together enclose the third mortise.

9. The floorboard of claim 8, wherein the second coupling structure further comprises a ninth plane; the ninth plane is configured to slope downwardly from an end of the second base surface away from the first base surface towards the direction away from the first mortise, and is connected to an end of the first plane away from the second plane; the second base surface, the ninth plane, the first plane, the second plane, the third plane and the fourth plane are configured to together enclose the third tenon; and in response to a case that two floorboards are connected, the second base surface of one of the two floorboards abuts against the third side portion of the other of the two floorboards, the ninth plane of one of the two floorboards abuts against the sixth inclined section of the other of the two floorboards, the first plane of one of the two floorboards abuts against the first straight section of the other of the two floorboards, the second plane of one of the two floorboards abuts against the first inclined section of the other of the two floorboards, the third plane of one of the two floorboards abuts against the second straight section of the other of the two floorboards, and the fourth plane of one of the two floorboards abuts against the second inclined section of the other of the two floorboards.